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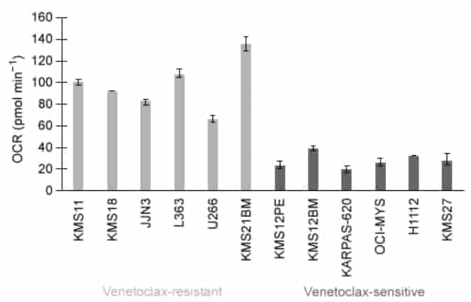

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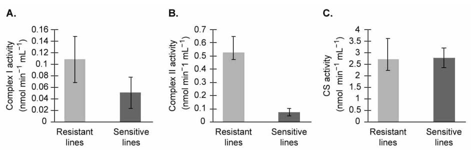


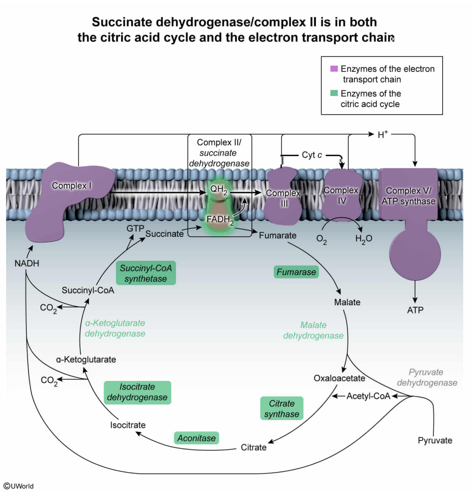
Figure 2 Enzyme activity of key respiratory enzymes. Basal activity from venetoclax-resistant and venetoclax-sensitive cell lines were compared for (A) Complex I (p = 0.008), (B) Complex II (p = 0.0007), and (C) citrate synthase (p = 0.2824).

One strategy to sensitize cancers to a particular treatment is to co-administer a second drug that brings the metabolic profile of a resistant phenotype closer to that of a sensitive one. Based on the passage, inhibition of which of the following enzymes or enzyme complexes is most likely to sensitize an otherwise resistant cell line to venetoclax?

- A. α-Ketoglutarate dehydrogenase [8%]
- ✔ B. Succinate dehydrogenase [65%]
- C. Malate dehydrogenase [6%]
- D. Pyruvate dehydrogenase [19%]

Correct
66% answered correctly

Explanation:



Metabolic pathways are highly **interconnected** through metabolic intermediates and the **enzymes** that interconvert them. The citric acid cycle (TCA cycle) and the electron transport chain (ETC) are examples of such interconnected pathways.

The proposed cancer treatment strategy assumes that the differences in ETC complex activities underlie venetoclax sensitivity differences. The passage states that the activities of Complexes I and II are inversely related to sensitivity. Thus, a reasonable strategy to increase sensitivity is to inhibit Complex I or II. **Complex II** of the ETC **is a multifunctional enzyme** that also functions in the TCA cycle, where it is **known as succinate dehydrogenase**. Complex II oxidizes succinate to fumarate, generating an $FADH_2$. The high-energy electrons from $FADH_2$ are then deposited into other carriers for further transport. Therefore, inhibition of **succinate dehydrogenase** (ie, Complex II) will likely increase venetoclax sensitivity.

(Choices A and C) α-Ketoglutarate dehydrogenase and malate dehydrogenase are both enzymes in the TCA cycle. Although both enzymes produce NADH that feeds into Complex I, inhibition of these enzymes does *not* affect Complex I directly—it does so only through substrate concentrations. According to the passage, only direct inhibition of Complex I or II has evidence to support their choice as targets that may increase venetoclax sensitivity.

(Choice D) Pyruvate dehydrogenase oxidatively decarboxylates pyruvate to produce acetyl-CoA and NADH. Although these products feed into the TCA cycle and ETC, only the activity of direct members of the ETC, not upstream pathways, has an effect on venetoclax sensitivity.

Educational objective:
The citric acid cycle and the electron transport chain are interconnected pathways central to aerobic respiration and energy metabolism. The citric acid cycle oxidizes acetyl-CoA to produce NADH and $FADH_2$, which then feed into the ETC. Succinate dehydrogenase, also known as Complex II of the ETC, oxidizes succinate to fumarate to make $FADH_2$ and then transfers the electrons from $FADH_2$ to subsequent acceptors for further transport.

Time spent:0 sec
Subject:
Biochemistry

QID:403185
System:
Metabolic Reactions

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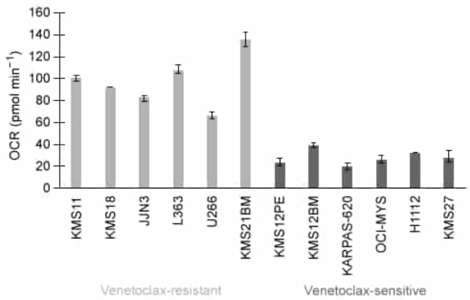

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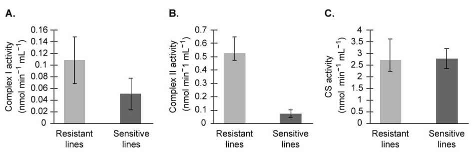


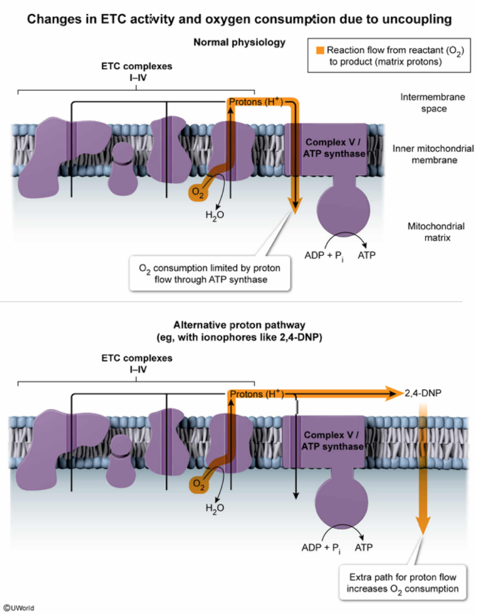
Figure 2 Enzyme activity of key respiratory enzymes. Basal activity from venetoclax-resistant and venetoclax-sensitive cell lines were compared for (A) Complex I (p = 0.008), (B) Complex II (p = 0.0007), and (C) citrate synthase (p = 0.2824).

2,4-Dinitrophenol (2,4-DNP) is a chemical that has historically been used for weight loss due to its ability to inhibit ATP synthesis. 2,4-DNP is a proton ionophore that inserts into the inner mitochondrial membrane, allowing protons to passively move from one side to the other. When measuring OCR, scientists may add an ionophore like 2,4-DNP as a control, expecting to see:

- ✔ A. increased oxygen consumption as electron transport chain activity increases to help compensate for the decreased ATP production. [52%]
- B. increased oxygen consumption as electron transport chain activity decreases to help increase NADH levels. [5%]
- C. decreased oxygen consumption as electron transport chain activity increases to help compensate for the decreased ATP production. [12%]
- D. Decreased oxygen consumption as electron transport chain activity decreases to help increase NADH levels. [29%]

Correct
52% answered correctly

Explanation:



In aerobic respiration, oxygen acts as the final electron acceptor in a series of redox reactions that oxidize fuels such as glucose and fatty acids to generate ATP. In this process, reduced electron carriers such as NADH and $FADH_2$ deposit their electrons into the electron transport chain. As these electrons move from complex to complex within the ETC, **protons are pumped** to the intermembrane space of the mitochondria, generating a proton gradient. The energy stored by this gradient is called the **proton-motive force**. ATP synthesis is normally **coupled** to the ETC because ATP synthase (also known as Complex V) uses the movement of protons down their electrochemical gradient back into the mitochondrial matrix as the force that powers ATP synthesis.

In this question, 2,4-DNP provides an alternative path to dissipate the proton gradient in addition to ATP synthase. This weakens the proton-motive force and also removes a product of the electron transport chain: pumped protons. Via Le Châtelier's principle, this causes a rightward shift to restore equilibrium. The result is increased pumping of protons to the intermembrane space (ie, increased ETC activity) and increased consumption of the reactants, including oxygen (ie, increased OCR), which is the final electron acceptor.

(Choices B and D) Although ATP synthesis is inhibited by 2,4-DNP, ETC activity *increase*s.

(Choice C) With increased activity of the ETC, there is both increased product formation and reactant consumption. Therefore, the OCR will increase.

Educational objective:
The electron transport chain and ATP synthesis are linked through the coupling of the proton gradient. The ETC generates a proton gradient, and ATP synthase uses this proton-motive force to power oxidative phosphorylation and the generation of ATP. Decoupling the ETC from ATP synthesis results in increased oxygen consumption and decreased ATP synthesis.

Time spent:0 sec
Subject:
Biochemistry

QID:403186
System:
Metabolic Reactions

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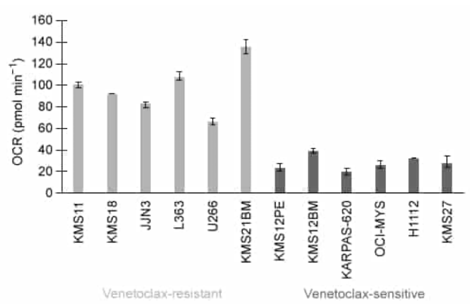

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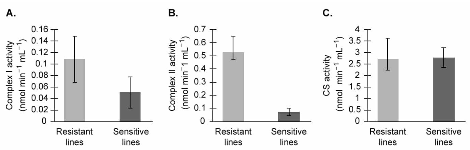


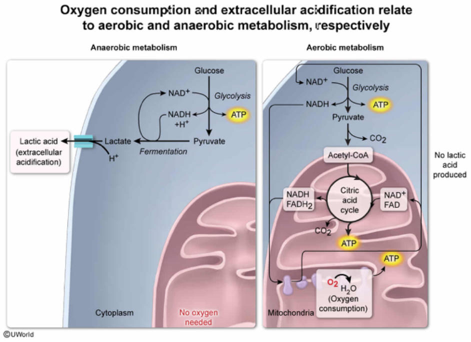
Figure 2 Enzyme activity of key respiratory enzymes. Basal activity from venetoclax-resistant and venetoclax-sensitive cell lines were compared for (A) Complex I (p = 0.008), (B) Complex II (p = 0.0007), and (C) citrate synthase (p = 0.2824).

Measurement of the extracellular acidification rate (ECAR) is often measured together with the OCR. Much of the extracellular acid measured comes from lactic acid, which itself comes from cytosolic lactate that is exported along with a proton. If venetoclax-sensitive cells were observed to have a lower-than-normal OCR/ECAR ratio, what would this imply about those cells?

- A. The cells have a high overall metabolic rate. [3%]
- B. The cells have a low overall metabolic rate. [14%]
- C. The cells prefer aerobic over anaerobic metabolism. [11%]
- ✔ D. The cells prefer anaerobic over aerobic metabolism. [70%]

Correct
69% answered correctly

Explanation:



Mammalian cells can release the energy from glucose in two ways: **aerobically** and **anaerobically**. Under aerobic (plentiful oxygen) conditions, pyruvate from glycolysis is transported to the mitochondria, where it becomes further oxidized through the citric acid cycle. The NADH and $FADH_2$ produced there then enter the electron transport chain, where oxygen is consumed during its role as the final electron acceptor. In anaerobic conditions, continual glycolysis results in the accumulation of pyruvate and NADH and the depletion of NAD^+. Fermentation allows for regeneration of NAD^+ stores by reacting pyruvate and NADH together to produce lactate.

The **oxygen consumption rate (OCR)** is a measurement of **aerobic metabolic activity**. The question states that the extracellular acid measured comes from lactic acid (ie, lactate and a proton), so the **extracellular acidification rate (ECAR)** is a measure of **anaerobic metabolic activity**. Taken together, the OCR/ECAR ratio measures the relative activity of aerobic to anaerobic metabolism. A low ratio indicates low aerobic activity relative to anaerobic (ie, that anaerobic activity is higher than aerobic activity). Therefore, if venetoclax-sensitive cells display a low OCR/ECAR ratio, this would imply that the cells prefer *an*aerobic metabolism.

(Choices A and B) Increases in a cell's overall metabolism may result in increased OCR, increased ECAR, or both. Similarly, decreases in overall metabolism may result in decreased OCR, ECAR, or both. Consequently, the *ratio* of OCR to ECAR cannot provide any information about overall metabolic rate.

(Choice C) A preference toward aerobic respiration would result in high oxygen consumption and a high OCR. In this case, a *high* OCR/ECAR ratio would be expected.

Educational objective:
Aerobic respiration consumes oxygen to completely oxidize glucose for efficient ATP production. Without oxygen, anaerobic metabolism reduces pyruvate to lactate while also regenerating NAD^+ to allow for continued partial oxidation of additional glucose. Oxygen consumption and lactic acid production can be used as measures of aerobic and anaerobic metabolism, respectively.

Time spent: 0 sec	**QID**:403187
Subject: Biochemistry	**System:** Metabolic Reactions

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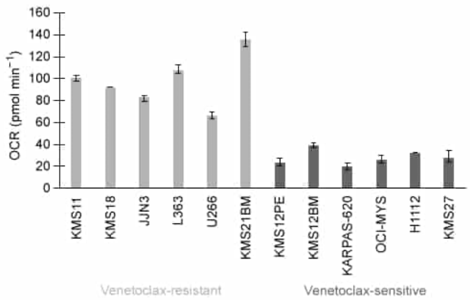

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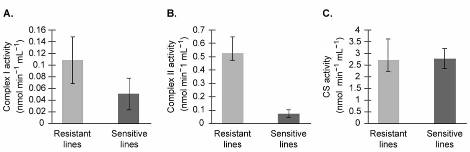

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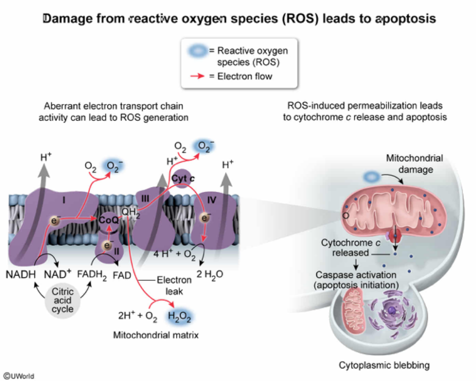
Given that venetoclax induces death on cells already primed for apoptosis, which of the following characteristics most likely primes a cell for a response to venetoclax?

- ○ A. Low rates of oxidative phosphorylation cause low ATP levels, priming cells for the failure to maintain integrity of the Na/K pump and rupture of the plasma membrane. [33%]
- ✔ ● B. Low ETC activity causes baseline oxidative stress, priming cells for the permeabilization of the outer mitochondrial membrane. [47%]
- ○ C. Higher levels of aerobic respiration completely oxidize most fuel molecules to CO_2, leaving little biomass for maintenance or cell division. [5%]
- ○ D. Higher activity levels of Complexes I and II deplete NADH, $FADH_2$, and NADPH levels, which are used to prevent oxidative damage. [13%]

Correct
46% answered correctly

Explanation:



Apoptosis is a programmed, noninflammatory process resulting in cell death. **Oxidative stress**, often caused by reactive oxygen species (ROS), can lead to **permeabilization of the mitochondrial membranes**, allowing for the **escape of cytochrome *c*** from the intermembrane space. This activates a cascade that eventually leads to the **activation of degradative enzymes** such as caspases and ultimately to cell death.

As electrons are shuttled across the electron transport chain (ETC), some electrons can leak out prematurely, attaching to oxygen in an uncontrolled manner that leads to the formation of ROS. Events that slow down the ETC, such as hypoxia or low Complex I or II activity, increase the opportunity for electrons to leak out because the normal pathway toward reduction of oxygen at complex IV has been obstructed. In cases where mitochondria are already stressed by ROS, venetoclax can more easily provide the final push toward activation of apoptosis.

(Choice A) The rupture of the plasma membrane would indicate necrosis, not apoptosis.

(Choices C and D) Higher levels of aerobic respiration and Complex I and II activity are characteristics of venetoclax-*resistant* cell lines that are *not* primed for apoptosis.

Educational objective:
Altered function of the electron transport chain can lead to generation of reactive oxygen species (ROS). ROS causes oxidative stress to mitochondria, which in turn may result in the release of cytochrome *c*. Cytochrome *c* release leads to apoptosis.

Time spent:0 sec
Subject:
Biochemistry

QID:403188
System:
Metabolic Reactions